

Ayush Kumar

 Ayush Kumar |  ayush-244 |  Ayush Kumar |  +919162573098  ayushku244@gmail.com

SUMMARY

B.Tech CSE student with strong foundations in Data Structures, Algorithms, and Object-Oriented Programming, experienced in building scalable full-stack applications and backend systems. Passionate about solving real-world problems using technology and interested in applying software engineering in financial services and large-scale systems.

WORK EXPERIENCE

Summer Research Intern – SRM University-AP

Jun 2025 – Jul 2025

- Built end-to-end ML system using Gradient Boosting to predict credit scores from 50,000+ financial records.
- Deployed interactive Streamlit dashboard with 100ms inference time for real-time decision support.
- Designed modular pipeline (ingestion → features → training → serialization) for production reliability.

Core Team Member – Google Developer Groups (GDG)

Aug 2025 – Present

- Tracked registration data across 6 events (800+ attendees) using Excel, improving outreach by 35%.
- Increased participant engagement by 30% through coordinated workshops across 4 technical sessions.
- Led campaigns reaching 3,500+ students, increasing community participation by 40%.

PROJECTS

MediBridge AI

[Link](#)

React, Node.js, Express.js, MongoDB, FastAPI, Python, ChromaDB

- Architected a healthcare SaaS using React 18, Node.js, and MongoDB with 4 role-based access levels.
- Engineered a RAG-based clinical AI service using FastAPI, ChromaDB, and DeepSeek LLM with 6 AI endpoints.
- Implemented real-time Socket.IO updates and AI-powered hospital matching across 3 Dockerized microservices.

AI-Based Peer Behavior Similarity Detector

[Link](#)

Python, Scikit-learn, NetworkX, NumPy, Pandas, Streamlit

- Built a system to detect anomalous behavior patterns in online exams using similarity metrics and graph-based analysis.
- Computed composite suspicion scores using cosine similarity (behavior vectors) and Jaccard similarity (answer overlap).
- Implemented graph-based collusion detection with NetworkX and deployed results using Streamlit.

Collaborative Class – Academic Collaboration & Analytics Platform

[Link](#)

Node.js, Express.js, MongoDB, Mongoose, JWT, Socket.IO

- Architected a role-based academic platform with assignment management, peer mentorship, and section-level access control.
- Implemented secure JWT authentication, lifecycle validation, and real-time doubt discussions using Socket.IO.
- Built analytics dashboards using MongoDB pipelines to track performance, submissions, and risks.

EDUCATION

B.Tech (CSE) at **SRM University AP** (CGPA: 8.85/10.0)

2023 - present

Class 12th CBSE Board [Cambridge Senior Sec. School, Buxar (Bihar)]

2022

Class 10th CBSE Board [Foundation School Buxar (Bihar)]

2020

SKILLS

Languages

Python, Java, JavaScript, TypeScript, C++, C, SQL

Frontend

React.js, Next.js, TypeScript, HTML, CSS, Tailwind CSS, Bootstrap

Backend

Node.js, Express.js, FastAPI, Flask, REST APIs, JWT Authentication, Socket.IO

AI / Machine Learning

Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, TensorFlow, PyTorch, Scikit-learn, CNNs, Transfer Learning

AI Tools & Frameworks

LangChain, ChromaDB, Qdrant, Sentence Transformers, Hugging Face Transformers, OpenRouter, DeepSeek LLM

Databases

MongoDB, MySQL

Data Science

Pandas, NumPy, Data Preprocessing, Feature Engineering, Data Visualization, Model Evaluation

Tools & Platforms

Git, GitHub, Docker, Linux, VS Code, Jupyter Notebook, Google Colab, Streamlit

CERTIFICATIONS

• **Oracle Cloud Infrastructure 2025 Certified Data Science Professional** – Oracle (2025)

[Link](#)

• **Oracle Certified Professional: Java SE 17 Developer** – Oracle (2026)

[Link](#)

• **MongoDB Certified Associate Developer (Node.js)** – MongoDB (2026)

[Link](#)